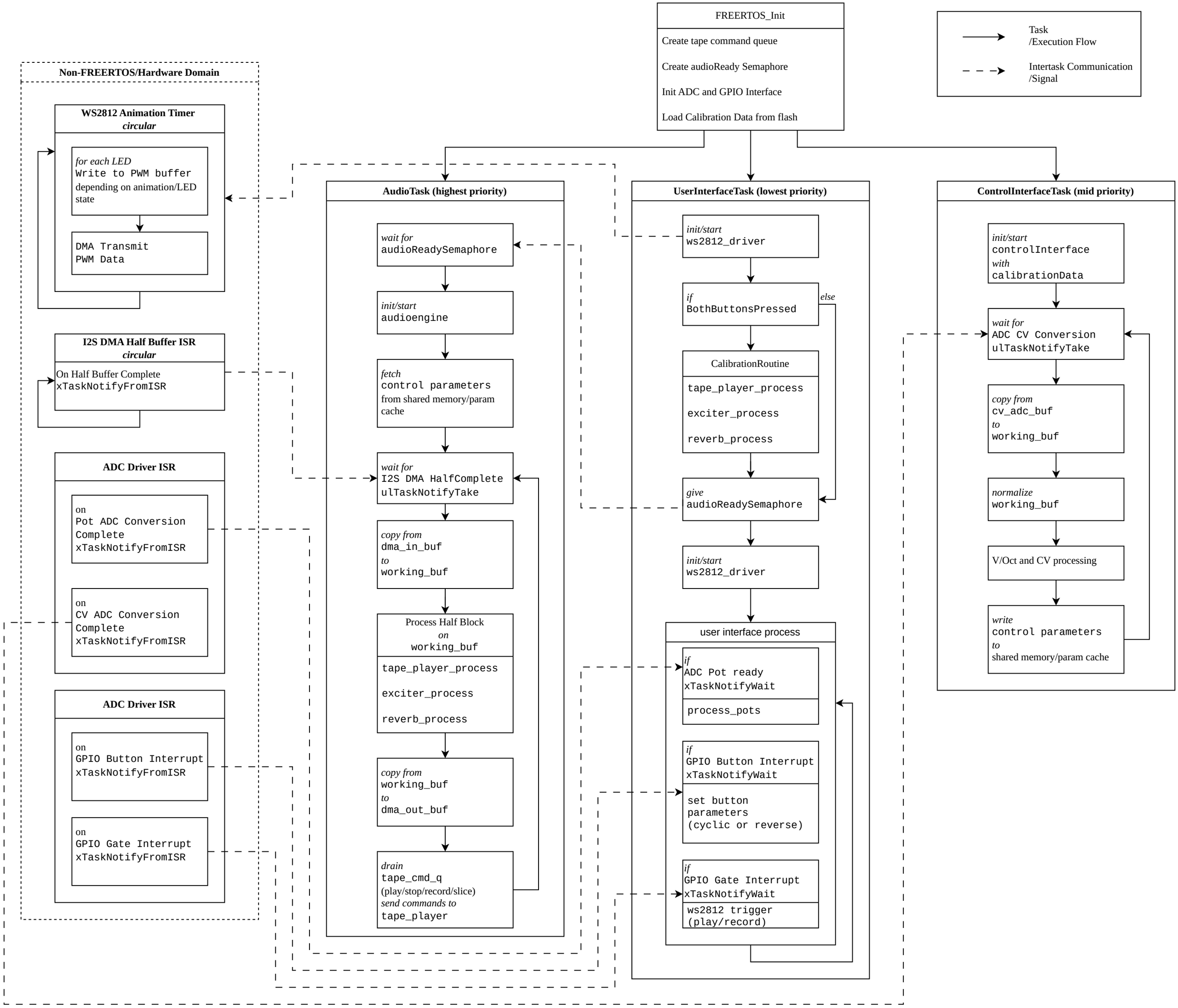
FREERTOS_Init
Create tape command queue
Create audioReady Semaphore
Init ADC and GPIO Interface
Load Calibration Data from flash
Task /Execution Flow
Intertask Communication /Signal
Non-FREERTOS/Hardware Domain
WS2812 Animation Timer
circular
for each LED
Write to PWM buffer
depending on animation/LED state
DMA Transmit PWM Data
I2S DMA Half Buffer ISR
circular
On Half Buffer Complete
xTaskNotifyFromISR
ADC Driver ISR
on
Pot ADC Conversion Complete
xTaskNotifyFromISR
on
CV ADC Conversion Complete
xTaskNotifyFromISR
ADC Driver ISR
on
GPIO Button Interrupt
xTaskNotifyFromISR
on
GPIO Gate Interrupt
xTaskNotifyFromISR
AudioTask (highest priority)
wait for
audioReadySemaphore
init/start
audioengine
fetch
control parameters
from shared memory/param cache
wait for
I2S DMA HalfComplete
ulTaskNotifyTake
copy from
dma_in_buf
to
working_buf
Process Half Block
on
working_buf
tape_player_process
exciter_process
reverb_process
copy from
working_buf
to
dma_out_buf
drain
tape_cmd_q
(play/stop/record/slice)
send commands to
tape_player
UserInterfaceTask (lowest priority)
init/start
ws2812_driver
if
BothButtonsPressed
else
CalibrationRoutine
tape_player_process
exciter_process
reverb_process
give
audioReadySemaphore
init/start
ws2812_driver
user interface process
if
ADC Pot ready
xTaskNotifyWait
process_pots
if
GPIO Button Interrupt
xTaskNotifyWait
set button parameters
(cyclic or reverse)
if
GPIO Gate Interrupt
xTaskNotifyWait
ws2812 trigger
(play/record)
ControlInterfaceTask (mid priority)
init/start
controlInterface
with
calibrationData
wait for
ADC CV Conversion
ulTaskNotifyTake
copy from
cv_adc_buf
to
working_buf
normalize
working_buf
V/Oct and CV processing
write
control parameters
to
shared memory/param cache